

KRISHNA LAKSHETA

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Yogyakarta, 55284, Indonesia

Master of Engineering student specializing in Power Electronics and Control Systems. Extensive experience in SST modeling (MMC/DAB) and HIL simulation using RT Box and Typhoon HIL, with control prototyping via PLECS Coder on TI Launchpad. Current research focuses on fault-tolerant control for DAB converters. Experienced in integrating machine learning with power electronic systems.

Education Level

Universitas Gadjah Mada - Indonesia

Aug 2020 - Aug 2024

Bachelor of Electrical Engineering, 3.71/4.00

Universitas Gadjah Mada - Indonesia

Feb 2025 - Dec 2026 (Expected)

Master of Electrical Engineering, 3.93/4.00

Research & Project Experience

Control & Simulation Researcher (MMC-SST Integration)

Aug 2025 - Present

Universitas Gadjah Mada – PT PLN (Persero) Collaboration

- Developing full-scale Solid-state Transformer (SST) simulations in Typhoon HIL, integrating Modular Multilevel Converter (MMC) rectifier, DAB, and inverter stages.
- Simplifying complex converter models to optimize computational resource usage while maintaining high-fidelity simulation for real-time execution.
- Implementing high-level control architectures utilizing a PLC as the master controller.

Graduate Researcher (DAB Fault-Tolerant Control)

Feb 2025 - Present

Universitas Gadjah Mada

- Developing Fault-Tolerant Control (FTC) strategies for Dual Active Bridge (DAB) converters to enhance system reliability and resilience.
- Researching and simulating advanced topologies to mitigate the impact of hardware faults in mission-critical power electronics applications.

Power Electronics Control Researcher (SST & DAB Hardware Testing)

Nov 2024 - Jul 2025

Universitas Gadjah Mada – PT PLN (Persero) Collaboration

- Developed Solid State Transformer (SST) integration—Modular Multilevel Converter (MMC) rectifier, DAB, and MMC inverter stages—including the design of startup and coordination schemes in PLECS.
- Validated DAB control algorithms (startup and closed-loop) on a physical 7kW prototype using the TI F28379D microcontroller.

Embedded Power Electronics & HIL Researcher (MMC Systems)

Nov 2023 - Jul 2024

Universitas Gadjah Mada – PT PLN (Persero) Collaboration

- Programmed Modular Multilevel Converters (MMC) control logic for TI C2000 microcontrollers—F280049C, F28069M, and F28379D—utilizing PLECS Coder for automated code generation.
- Validated Solid State Transformer (SST) real-time performance via high-fidelity Typhoon HIL 604 testing.
- Synchronized complex signal routing across multiple Digital Signal Processors (DSPs) to ensure precise regulation and converter stability.

Embedded Power Electronics & HIL Researcher (DAB Systems)

Feb 2023 - Jun 2023

Universitas Gadjah Mada – PT PLN (Persero) Collaboration

- Programmed Dual-Active Bridge (DAB) control logic for TI F28069M microcontrollers using PLECS Coder.
- Validated control prototypes through real-time simulation via Typhoon HIL 604 to ensure converter stability and design accuracy.

Professional & Academic Experience

Machine Learning Student

Aug 2023 - Dec 2023

Bangkit Academy

- Developed a Solar Intensity Prediction System for solar panel forecasting using Python (TensorFlow, NumPy, Pandas) for regression-based time-series modeling.

Electronics Division

Jul 2021 - Jul 2023

Gajah Mada Robotic Team (Fukuro)

- Led the electronics team for autonomous soccer robots in the KRSBI (Indonesian Soccer Robot Competition).
- Engineered high-voltage kicker boards with solenoid valves for precise robot actuation.
- Managed PCB design and component selection for autonomous robots to ensure reliability in high-interference environments.

PCB Designer

Jun 2022 - Jul 2022

PT Helber Elektronika Indonesia

- Designed and developed an integrated PCB merging the Arduino Uno and ESP 12-E to create a unified board for wireless embedded applications.

Academic Tutor

Aug 2021 - Jun 2022

Universitas Gajah Mada

- **Complex Variable Analysis:** Mentored weekly classes on analytic functions, Cauchy's Theorem, Cauchy's Integral Formula, and the application of Taylor, Laurent, and Fourier series.
- **Linear Algebra:** Conducted weekly classes on Gauss- Jordan elimination, LU factorization, Singular Value Decomposition (SVD), and eigenvalue decomposition.

Skills

- **Simulation:** PLECS, Typhoon HIL, LTspice.
- **Hardware-in-the-Loop (HIL):** Typhoon HIL 604, Plexim RT Box.
- **Embedded Systems & Control:** Texas Instruments (TI) C2000 microcontrollers (F28379D, F28069M, F280049C).
- **Programming & Data:** Python (TensorFlow, NumPy, Pandas), Markdown, LaTeX.
- **Hardware Design:** PCB Design, High- Voltage (HV) circuit prototyping, solenoid valve integration, power electronics hardware validation (up to 7kW).
- **Languages:** Indonesian (Native), English (Intermediate).

Publications

- **Krishna Laksheta, Y. F. Sidik, and F. D. Wijaya, "Buck Converter Parameters Design Using Artificial Intelligence-Based Genetic Algorithm," 2024 16th International Conference on Information Technology and Electrical Engineering (ICITEE), 2024. doi: [10.1109/ICITEE62483.2024.10808195](https://doi.org/10.1109/ICITEE62483.2024.10808195).**
- **L. N. Wijaya, Y. F. Sidik, F. D. Wijaya, and Krishna Laksheta, "Voltage Control Using Long-Short-Term Memory (LSTM) for Dual-Active-Bridge (DAB) Converter," 2025 17th International Conference on Information Technology and Electrical Engineering (ICITEE), 2025. doi: [10.1109/ICITEE66631.2025.11338242](https://doi.org/10.1109/ICITEE66631.2025.11338242).**
- **M. Ahmad, M. Mudarris, E. Firmansyah, F. D. Wijaya, Y. F. Sidik, Krishna Laksheta, and L. N. Wijaya, "Jitter Clock Effect Analysis on Modular Dual Active Bridge (MDAB) Converter," 2023 International Conference on Advanced Mechatronics, Intelligent Manufacture and Industrial Automation (ICAMIMIA), 2023. doi: [10.1109/ICAMIMIA60881.2023.10427750](https://doi.org/10.1109/ICAMIMIA60881.2023.10427750).**